

JEE MAIN REVISION MASTER: AREA UNDER THE CURVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 170 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Area Under The Curves — Core Principles & Governing Laws	127 PYQs	Core Principle
Very High	Area Under The Curves — Standard Calculations & Problem Archetypes	43 PYQs	Quantitative Analysis
Standard	Area Under The Curves — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Area Under The Curves — Core Principles & Governing Laws	$x^2 = x + 2, \quad \frac{1}{k}=4b, k>0, \quad x^2+4x+2 \leq y \leq x+2 $
Area Under The Curves — Standard Calculations & Problem Archetypes	$x^2+y^2=4, \quad \{(x,y):1-2x \leq y \leq 4-x^2, x \geq 0, y \geq 0\}, \quad f(x)=\max\{x^2, 1+[x]\}$
Area Under The Curves — Applied Cases, Reactions & Traps	$\{(x,y):0 \leq y \leq 6-x, y^2 \geq 4x-3, x \geq 0\}, \quad \{(x,y):0 \leq y \leq 6-x, y^2 \geq 4x-3, x \geq 0\}.$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Area Under The Curves — Core Principles & Governing Laws (Very High Yield — 127 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Area Under The Curves.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Area Under The Curves.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Area Under The Curves PYQs.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $f:(1,\infty) \rightarrow R$ be a function defined as $f(x)=(x-1/x+1)$. Let $f^{i+1}(x)=f(f^i(x))$, $i=1,2,\dots,25$, where $f^1(x)=f(x)$." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 27, 1 \leq y \leq x^2$ is equal to :..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "The area of the region $A=(x,y):4x^2+y^2 \leq 8$ and $y^2 \geq 4x$ is:..." [Ans: N/A]

3.2 Area Under The Curves — Standard Calculations & Problem Archetypes (Very High Yield — 43 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Area Under The Curves.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The area of the region $R=(x,y):xy \leq 8, 1 \leq y \leq x^2, x \geq 0$ is..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Let $f:R \rightarrow R$ be a function such that $f(x)+3f((\pi/2)-x)=\sin x, x \in R$. Let the maximum value of f on R be α" [Ans: 16]
- [2026 Evening Shift] Q#2 (MCQ): "The area of the region $(x,y):x^2-8x \leq y \leq -x$ is :..." [Ans: A]

3.3 Area Under The Curves — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Area Under The Curves.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#3 (MCQ): "The area of the region $(x,y):0 \leq y \leq 6-x, y^2 \geq 4x-3, x \geq 0$ is :..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Evening Shift (Q#1)

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x, x \in \mathbb{R}$. Let the maximum value of f on $\mathbb{R}$ be α . If the area of the region bounded by the curves $g(x) = x^2$ and $h(x) = \beta x^3, \beta > 0$, is α^2 , then $30\beta^3$ is equal to ____.

Correct Answer: 16

Solution Analysis: $f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x \dots (1)$ Now replace x by $\left(\frac{\pi}{2} - x\right)$ in (1). This is a standard NCERT step to get a second equation in the same two terms. $\Rightarrow f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x \dots (2)$ Now we solve the two linear equations (1) and (2) to find $f(x)$. Multiply (1) by 3 : $3f(x) + 9f\left(\frac{\pi}{2} - x\right) = 3\sin x \dots (3)$ Subtract (2) from (3): $(3f(x) - 3f(x)) + (9 - 1)f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$ $8f\left(\frac{\pi}{2} - x\right) = 3\sin x - \cos x$ Now put $x \rightarrow \left(\frac{\pi}{2} - x\right)$ in this result to get $f(x)$: $8f(x) = 3\cos x - \sin x$ $f(x) = \frac{1}{8}(3\cos x - \sin x)$ To find the maximum value,

Benchmark Problem 2: JEE Main 2026 (Online) 2nd April Evening Shift (Q#2)

If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A , then $3(A + 6\log_e(3))$ is equal to _____.

Correct Answer: 24

Solution Analysis: $16x^2 - 9y^2 = 144$ $16x^2 - (8x - 24)^2 = 144$ $16x^2 - 64(x - 3)^2 = 144 \rightarrow x^2 - 4(x - 3)^2 = 9$ $3x^2 - 24x + 45 \rightarrow x^2 - 8x + 15 = 0$ $x = 3, 5$ $\text{Area} = \int_3^5 \sqrt{\frac{16x^2}{9} - 144} - (1/2) \cdot 2 \cdot (16/3) = (4/3) \int_3^5 \sqrt{(x^2) - 9} - (16/3) = (4/3) \left(\frac{x}{2} \sqrt{(x^2) - 9} - (9/2) \log_e(x + \sqrt{(x^2) - 9}) \right) \Big|_3^5 - (16/3)$ $= (4/3) \left(\frac{5}{2} \cdot 4 - (9/2) \log_e(9 - (3/2)) \cdot 0 + (9/2) \log_e(3) - (16/3) \right)$ $\text{Area} = 8 - 6\log_e(3) = A \therefore A + 6\log_e(3) = 8 \Rightarrow 3(A + 6\log_e(3)) = 24$